

The Locked Foundation of Resolution Geometry

Exact Observation, Singular Diagnostics, Observable Transport, and the Gated Construction of Causal Cones

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Abstract

This paper records the part of Resolution Geometry that survived a sequence of hostile symbolic and numerical attacks. The starting point is an observation map $F : X \rightarrow Y$. The exact kernel pair R_F and quotient Q_F are forced. After a declared differentiable statistical lift (F, Σ) with $\Sigma \succ 0$, the pullback form $g_F = F^*(\Sigma^{-1})$ is positive semidefinite and has radical $\ker dF$. On constant-rank strata this radical equals the tangent to the exact fibre. At singular strata it may contain linearization relicts. A multiscale profile $\rho_x(v, r)$ can witness that a differential null direction is not an exact fibre direction, but finite numerical silence cannot certify exact nullness in unrestricted C^∞ models.

The principal positive result lies beyond this differential layer. For admissible interventions a with observable response operators T_a , projectability and composability define a representation of the resolved intervention category. The closure defect

$$\Delta_F(a, b) = \frac{\|T_{boa}^{\text{true}} - T_b T_a\|}{1 + \|T_{boa}^{\text{true}}\|}$$

is an operational obstruction to resolved dynamical sufficiency. In the tested block-linear model, decoupled null variables gave mean defect 0.0000, while null variables carrying two-step memory gave mean defect 0.2394 under seed 20260912. The theorem is not that every nonzero defect has one unique hidden cause. The theorem is that exact projectable composition yields a functor, while systematic failure of composition proves that the chosen resolved state and operator assignment do not close.

Causal and geometric reconstruction is therefore placed behind a firewall. Static relict data do not determine a response, memory kernel, causal orientation, affine connection, or conformal metric. A causal cone may be computed only from a separately established directed intervention-response relation that descends to Q_F , is stable under the declared resolution protocol, and passes a Lorentzian quadratic-cone gate. Under the usual causal and topological hypotheses, the resulting order can determine a conformal class, but not an absolute scale. The paper states the final theorem and corollary ladder, gives explicit cone-construction algorithms and figures, records all executed batteries and seeds, and separates analytic conclusions from numerical reproducibility evidence.

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1 Purpose and claim discipline

The purpose is not to present a differential pullback as a new result. Pullbacks, constant-rank fibres, causal orders, functors, and holonomy are established mathematical objects. The novel foundation is the *typed closure*: each object is assigned to the earliest layer at which it is defined, every type crossing receives an explicit gate, and the experimentally accessible obstruction to dynamic descent is isolated as Δ_F .

Locked result

The final survivor is a chain of exact observation, declared local distinguishability, one-sided singular diagnosis, and operational response composition. Geometry is downstream. It is reconstructed only if intervention support, quotient projectability, cone regularity, and an appropriate realization theorem all pass.

We use four labels throughout.

- Forced.** A consequence of the stated input by proof.
- Declared.** Additional structure that is selected or measured, not reconstructed from a lower layer.
- Gated.** A consequence valid only under an explicit hypothesis.
- Open or empirical.** A proposed compatibility law requiring independent attack.

2 The final K-ladder

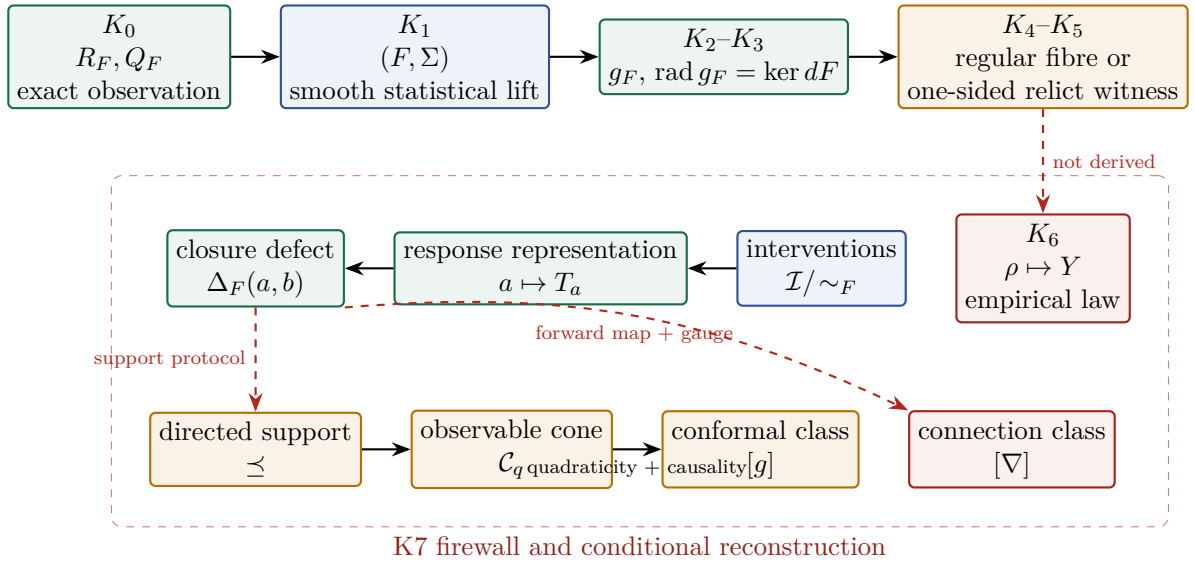


Figure 1: The final typed ladder. Solid arrows are forced once their displayed inputs and gates are supplied. Dashed arrows require independent empirical or reconstruction laws.

$K_0 : R_F, Q_F$	exact extensional layer, forced by F ,
$K_1 : (F, \Sigma)$	declared differentiable statistical lift,
$K_2 : g_F = F^*(\Sigma^{-1})$	pullback form, forced relative to K_1 ,
$K_3 : \text{rad } g_F = \ker dF$	differential null sector, forced,
$K_4 : \ker dF = T(F^{-1})$	constant-rank gate,
$K_5 : \rho_x(v, r) > 0$	one-sided witness against exact nullness,
$K_6 : \rho \mapsto Y$	independent empirical compatibility law,
$K_7 : \mathcal{I}/\sim_F \rightarrow \text{End}(\mathcal{O})$	projectable composable response representation.

The downstream objects K , $\preceq$, $[g]$, and $[\nabla]$ are not outputs of static relict data. They require separately defined forward maps and identifiability conditions.

3 Exact observation and local distinguishability, K0–K3

Theorem 3.1 (Exact observational root). *Let $F : X \rightarrow Y$. Then*

$$R_F = X \times_Y X = \{(x, x') : F(x) = F(x')\}$$

is an equivalence relation. The quotient $Q_F = X/R_F$ is canonically in bijection with $\text{Im } F$. Every map $h : X \rightarrow Z$ that is constant on F -fibres factors uniquely through the quotient map $\pi_F : X \rightarrow Q_F$.

Proof. Reflexivity, symmetry, and transitivity follow from equality in Y . Define $\bar{F}([x]) = F(x)$. This is well-defined and bijective onto $\text{Im } F$. If h is fibre-constant, define $\bar{h}([x]) = h(x)$. Well-definedness and uniqueness follow immediately. $\square$

Corollary 3.2 (Projectability criterion). *A construction on representatives descends to Q_F if and only if it is invariant under replacement by R_F -equivalent representatives in every argument on which it depends.*

Definition 3.3 (Declared statistical lift). Let M and Y be differentiable manifolds, let $F : M \rightarrow Y$ be differentiable, and let Σ_y be a positive-definite covariance form on observation space. Define

$$g_F(x) = F^*(\Sigma^{-1})_x, \quad g_F(v, w) = \Sigma_{F(x)}^{-1}(dF_x v, dF_x w).$$

Theorem 3.4 (Pullback form and radical). *For the declared lift above, g_F is positive semidefinite and*

$$\text{rad } g_F(x) = \ker dF_x.$$

Under an invertible observation-coordinate change $y' = Ay$ and parameter-coordinate change $x = Cx'$, with $\Sigma' = A\Sigma A^\top$ and $J' = AJC$, the matrix representation obeys

$$g'_F = C^\top g_F C.$$

Proof. Positive definiteness of Σ^{-1} gives

$$g_F(v, v) = \Sigma^{-1}(dFv, dFv) \geq 0,$$

with equality if and only if $dFv = 0$. Coordinate covariance follows by substitution:

$$J'^\top \Sigma'^{-1} J' = C^\top J^\top A^\top (A\Sigma A^\top)^{-1} AJC = C^\top J^\top \Sigma^{-1} JC.$$

$\square$

Audit record

The seed-20260912 battery executed 300 coordinate trials and obtained maximum relative covariance error 1.68×10^{-12} . For a rank-deficient Jacobian with one-dimensional kernel, the computed quadratic value on the numerical kernel was 2.00×10^{-31} . These values verify the implementation; the theorem itself is analytic.

Remark 3.5 (Fisher shadow, not exact root). The same kernel pair can be realized by differentiable maps with different derivatives. Thus g_F is canonical relative to (F, Σ) , not relative to R_F alone. The exact quotient and the local distinguishability form are different types.

4 Regular identification and singular relicts, K4–K5

Theorem 4.1 (Regular fibre identification). *If F has locally constant rank near x , then the fibre $F^{-1}(F(x))$ is a local submanifold and*

$$T_x F^{-1}(F(x)) = \ker dF_x = \text{rad } g_F(x).$$

Proof. This is the constant-rank theorem followed by Theorem 3.4. $\square$

Definition 4.2 (Linearization relict). At a point where a meaningful tangent or tangent cone to the exact fibre is available, the excess differential null sector is the discrepancy between $\ker dF_x$ and the exact fibre germ. If both are linear spaces, define

$$\text{Rel}_x^{(1)}(F) = \ker dF_x / T_x F^{-1}(F(x)).$$

Proposition 4.3 (Singular strictness). *For $F(x) = x^m$ with integer $m \geq 2$ at $x = 0$,*

$$\ker dF_0 = \mathbb{R}, \quad F^{-1}(F(0)) = \{0\}, \quad T_0 F^{-1}(0) = \{0\}.$$

Hence the first-order null direction is not exact ignorance.

Definition 4.4 (Finite-scale distinguishability profile). For $v \in \ker dF_x$, define

$$\rho_x(v, r) = \frac{\|F(x + rv) - F(x)\|_{\Sigma^{-1}}}{|r|},$$

whenever the chosen perturbation chart and norm are defined.

Theorem 4.5 (One-sided singular falsification). *If $v \in \ker dF_x$ and every neighbourhood of 0 contains an $s \neq 0$ with $F(x + sv) \neq F(x)$, then v is not an exact straight-line fibre direction. A finite experiment that observes $F(x + sv) \neq F(x)$ witnesses this failure. Failure to observe a difference at finitely many scales does not certify exact fibre membership in unrestricted C^∞ models.*

Proof. The witness statement is the negation of local constancy along the selected perturbation line. For non-certification, a smooth flat injective germ such as $\text{sgn}(x)e^{-1/x^2}$ has every derivative zero at the origin while differing from its value at the origin for every nonzero x . $\square$

5 The K6 interface: where a physical law must be posted

Definition 5.1 (Relict-response compatibility law). A K6 law consists of an independently measurable response Y , a declared summary or functional of ρ , matched controls, and a preregistered statement

$$Y = \mathcal{H}[\rho] \quad \text{or} \quad L[\rho] \leq Y \leq U[\rho].$$

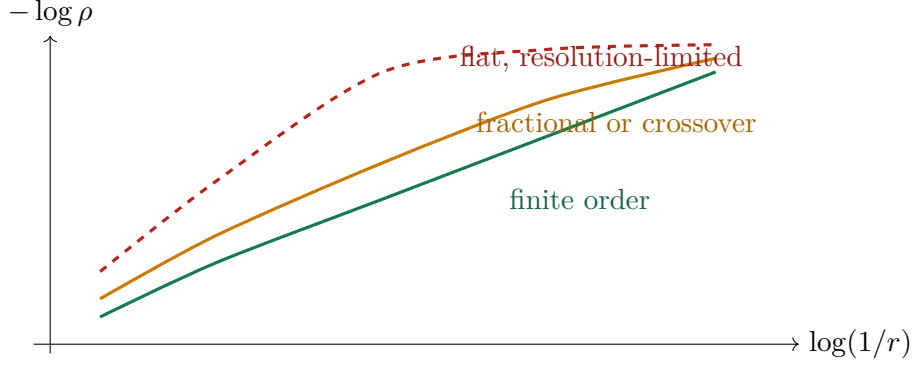


Figure 2: The profile is the robust object. A discrete integer exponent requires an additional finite-order analytic or tame gate.

Proposition 5.2 (K6 is not forced). *Static data (F, Σ, ρ) do not determine response gain, decay rate, phase, time orientation, memory horizon, or a memory kernel. Consequently $\rho \mapsto Y$ is an empirical compatibility law, not a consequence of K0–K5.*

Protocol 5.3 (Decisive K6 attack). Construct matched systems or regimes with controlled (F, Σ, ρ) . Measure Y independently. A deterministic proposed law fails if equal admissible ρ profiles yield different Y . An injective converse fails if different admissible ρ profiles yield indistinguishable Y . Statistical versions must preregister uncertainty, equivalence margins, and held-out evaluation.

6 K7: observable transport before geometry

6.1 Resolved interventions and true composition

Let Z be a full state space, $q : Z \rightarrow \mathcal{O}$ an observation or resolution map, and $i : \mathcal{O} \rightarrow Z$ a declared preparation of resolved states. For an intervention $a : Z \rightarrow Z$, define its one-step resolved response

$$T_a = q \circ a \circ i.$$

The true resolved response of a serial intervention is

$$T_{boa}^{\text{true}} = q \circ b \circ a \circ i.$$

In general,

$$T_{boa}^{\text{true}} \neq T_b \circ T_a,$$

because the intermediate preparation implicit in $T_b \circ T_a$ may erase unresolved state created by a .

Definition 6.1 (Normalized closure defect). For a selected operator norm, define

$$\Delta_F(a, b) = \frac{\|T_{boa}^{\text{true}} - T_b T_a\|}{1 + \|T_{boa}^{\text{true}}\|}.$$

Theorem 6.2 (Composability bridge). *Let $\mathcal{I}$ be a category of admissible interventions and let T_a be projectable observable response operators. If*

$$T_{\text{id}} = I, \quad T_{boa}^{\text{true}} = T_b T_a$$

for every composable pair, then $a \mapsto T_a$ is a functor

$$T : \mathcal{I} / \sim_F \longrightarrow \text{End}(\mathcal{O}).$$

Conversely, any witnessed pair with $\Delta_F(a, b) > 0$ proves that the chosen resolved state, preparation, and operator assignment do not form an exact compositional representation for that pair.

Proof. Identity and composition are precisely the functor axioms. Projectability makes the assignment well-defined on equivalence classes. A positive norm difference contradicts the composition axiom. $\square$

Remark 6.3 (Correct scope of “if and only if”). For one tested pair, $\Delta_F(a, b) = 0$ does not prove global sufficiency; accidental cancellation is possible. Global vanishing for all admissible pairs is equivalent to exact compositional closure of the selected representation. Interpreting that closure as a sufficient statistic for dynamics additionally requires that the intervention family is rich enough to probe the declared dynamics.

6.2 Block-linear memory model

Write the full state as $z = (r, m)$, with resolved component $r \in \mathbb{R}^{d_r}$ and unresolved component $m \in \mathbb{R}^{d_m}$. Let

$$A_a = \begin{pmatrix} R_a & U_a \\ V_a & M_a \end{pmatrix}, \quad A_b = \begin{pmatrix} R_b & U_b \\ V_b & M_b \end{pmatrix}.$$

Preparing $m = 0$ gives $T_a = R_a$ and $T_b = R_b$. The true two-step resolved block is

$$(A_b A_a)_{rr} = R_b R_a + U_b V_a.$$

Therefore

$$T_{bca}^{\text{true}} - T_b T_a = U_b V_a.$$

Proposition 6.4 (Two-step null-memory obstruction). *In the block-linear preparation model above,*

$$\Delta_F(a, b) = \frac{\|U_b V_a\|}{1 + \|R_b R_a + U_b V_a\|}.$$

The defect vanishes whenever the first intervention does not write into the unresolved sector ($V_a = 0$) or the second intervention does not read from it ($U_b = 0$). A nonzero defect certifies an unresolved two-step path $r \xrightarrow{V_a} m \xrightarrow{U_b} r$.

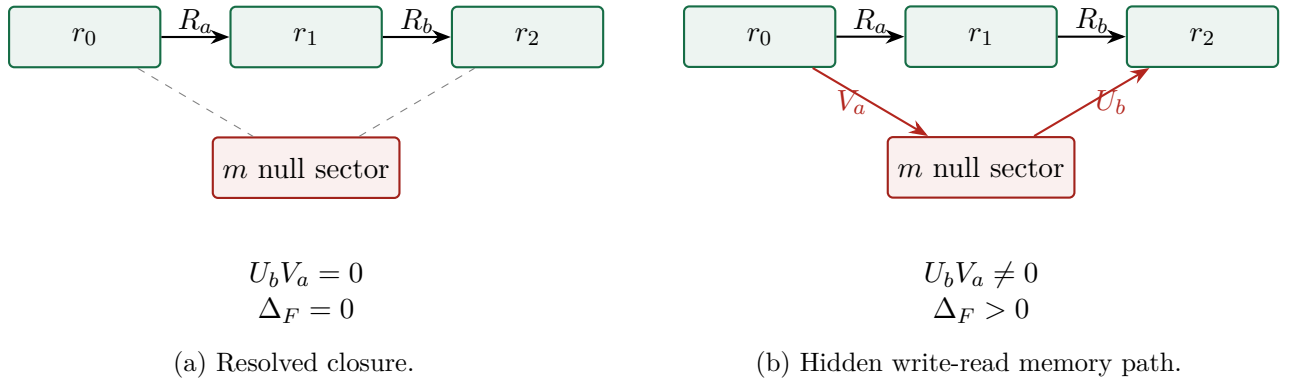


Figure 3: What the closure defect detects in the verified block-linear model.

Audit record

With $d_r = 3$, $d_m = 2$, 200 intervention pairs per regime, and seed 20260912, the executed battery gave mean $\Delta_F = 0.0000$ for block-diagonal interventions and mean $\Delta_F = 0.2394$ for coupling amplitude 0.6. This supports Proposition 6.4 and verifies the implementation. It does not prove that every physical defect is caused only by hidden memory; model misspecification, nonstationarity, noise bias, intervention dependence, or a wrong preparation map can also produce nonclosure.

6.3 Projectability and representative dependence

Theorem 6.5 (Dynamic descent criterion). *The response assignment descends to the observational quotient only if equivalent representatives produce equivalent observable responses under every admitted intervention and every admitted composition. If this fails, no response operator on Q_F is well-defined.*

Corollary 6.6 (Representative test). *A statistically significant dependence of T_a or $\Delta_F(a, b)$ on the chosen representative of one R_F -class falsifies dynamic projectability at the declared resolution.*

7 From response support to a causal preorder

Static distinguishability has no orientation. Direction enters through intervention and response.

Definition 7.1 (Directed response support). For quotient states or localized quotient regions $p, q \in Q_F$, write

$$p \rightsquigarrow_\eta q$$

if an allowed intervention localized at p produces a response at q exceeding the preregistered detection rule η .

Definition 7.2 (Operational reachability preorder). Let $\preceq_\eta$ be the reflexive transitive closure of $\rightsquigarrow_\eta$. Thus $p \preceq_\eta q$ means that q is reachable from p through a finite chain of detected directed responses.

Proposition 7.3 (What is forced by the construction). *For every fixed intervention class and detection rule, $\preceq_\eta$ is reflexive and transitive. Antisymmetry is not forced. If $p \preceq_\eta q$ and $q \preceq_\eta p$ for distinct points, the result is a preorder with a causal cycle or an unresolved equivalence, not automatically a partial order.*

Required gate

The operational order is meaningful only relative to: a declared intervention set; a response observable; a detection threshold or statistical decision rule; handling of latent common causes; temporal ordering of interventions and responses; stability under repetitions; and projectability through R_F . Changing any of these may change the order.

Algorithm 1 Operational preorder construction

Require: Quotient locations Q , intervention family $\mathcal{I}$, response statistic S , decision rule η .

- 1: **for all** $p, q \in Q$ **do**
 - 2: execute or estimate interventions localized at p
 - 3: set edge $p \rightsquigarrow_\eta q$ if $S(p \rightarrow q)$ passes η
 - 4: test representative invariance of every retained edge
 - 5: remove or stratify edges that fail projectability
 - 6: return the reflexive transitive closure $\preceq_\eta$
-

8 When and how a light cone can be computed

This section is constructive. It does not infer a light cone from g_F or ρ . It states the separate data and gates needed to estimate an observable cone from directed response support.

8.1 Local directional reachability

Let $q \in Q_F$ and suppose a local chart or experimentally controlled directional family $u \in S^{n-1}$ is available. For scale $r > 0$, define a directed response score

$$S_q(u, r) = \text{response at } q + ru \text{ caused by an admissible intervention at } q.$$

With decision rule η , define the finite-resolution directional set

$$\mathcal{C}_{q,r}^\eta = \{\lambda u : \lambda \geq 0, S_q(u, r) \text{ passes } \eta\}.$$

A limiting cone candidate requires stability over a declared scale window, not an unqualified $r \rightarrow 0$ extrapolation.

Definition 8.1 (Observable cone candidate). If the directional sets are stable under the preregistered scale and threshold protocol, define

$$\mathcal{C}_q^{\text{obs}} = \text{cl cone}\{u : u \text{ is repeatedly reachable from } q\}.$$

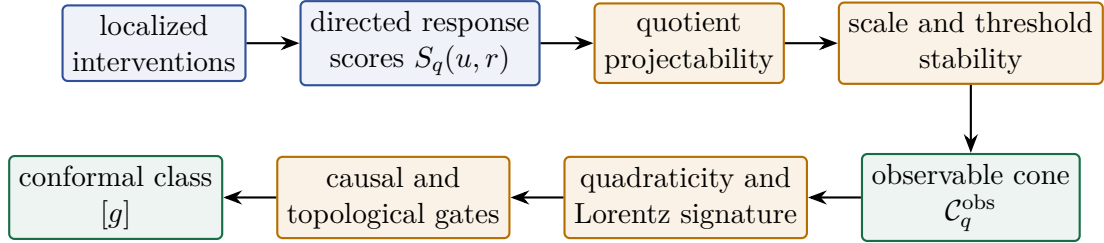


Figure 4: A computational workflow for an observable cone. The conformal metric is the last output, not the starting assumption.

8.2 Cone size

A cone has several distinct notions of size. They must not be conflated.

Definition 8.2 (Angular cone size). Given a declared auxiliary sampling measure μ on directions,

$$\Omega_q^{\eta,r} = \mu\{u \in S^{n-1} : S_q(u, r) \text{ passes } \eta\}.$$

This is a protocol-relative solid-angle fraction, not yet a metric invariant.

Definition 8.3 (Response-weighted cone size). For nonnegative weight w ,

$$W_q^{\eta,r} = \int_{S^{n-1}} w(S_q(u, r)) \mathbf{1}_\eta(S_q(u, r)) d\mu(u).$$

This records both reachable direction and response strength. Its value depends on the chosen response scale and weight.

Definition 8.4 (Projective cone shape). The projective cone is the set of rays $\mathbb{P}\mathcal{C}_q^{\text{obs}}$. It captures directional shape without radial scale. This is the appropriate object for conformal reconstruction because positive rescaling of a Lorentzian metric leaves the null rays unchanged.

Remark 8.5 (What “larger cone” can mean). A larger angular support, a larger weighted response, and a wider Lorentzian opening are different statements. Only comparisons made under the same chart-independent or calibrated protocol are meaningful. A conformal class fixes null directions but not an absolute magnitude for cone size.

8.3 Quadratic cone gate

Suppose boundary rays u_i have been estimated. A Lorentzian quadratic realization seeks a symmetric matrix G_q such that

$$u_i^\top G_q u_i \approx 0$$

for boundary directions, with one negative and $n - 1$ positive eigenvalues, up to overall nonzero scale.

Protocol 8.6 (Quadratic-cone fit). Vectorize the independent entries of G_q . Build a design matrix whose i th row contains the quadratic monomials of u_i . Compute the smallest singular vector as a candidate G_q . Accept only if: the residual is below the preregistered tolerance; the signature is Lorentzian; held-out boundary rays remain near-null; interior and exterior test rays have consistent signs; and the result is stable under resolution and representative changes.

Algorithm 2 Local conformal-cone reconstruction

Require: Projectable directed response data near q ; boundary classifier; dimension $n > 2$ for the intended causal realization theorem.

- 1: estimate reachable rays and boundary rays in several scale windows
 - 2: test threshold and scale stability
 - 3: fit symmetric G_q from $u_i^\top G_q u_i \approx 0$
 - 4: reject if the residual, held-out error, or Lorentzian signature gate fails
 - 5: orient future and past using the intervention time orientation
 - 6: return the ray $[G_q]$, not an absolute metric
-

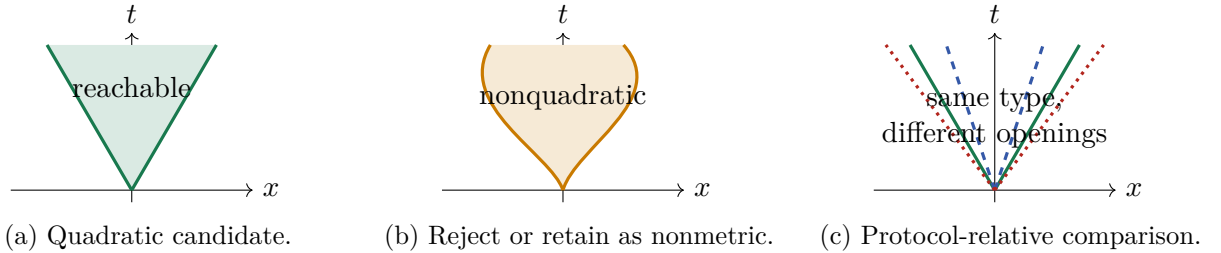


Figure 5: Cone interpretation. A cone boundary must pass the quadraticity gate before it is represented by a Lorentzian conformal metric.

Theorem 8.7 (Conditional conformal realization interface). *Assume an operational causal order has descended to a suitable spacetime manifold, the required causal and topological hypotheses hold, and the local directional boundaries are represented by Lorentzian quadratic cones. Then the causal order determines the conformal metric class under the hypotheses of the relevant Hawking–King–McCarthy–Malament type result. The absolute conformal factor is not determined by the order alone.*

Non-claim 8.8. Theorem 8.7 is an interface to established causal reconstruction mathematics. It is not derived from ρ , g_F , or Δ_F alone. The operational work is to establish the order, descent, topology, and quadraticity hypotheses.

9 Loop defects, holonomy, and the connection window

For a closed resolved intervention word $\gamma = a_n \circ \cdots \circ a_1$, define the observable loop operator

$$T_\gamma = T_{a_n} \cdots T_{a_1}$$

when the response representation closes, and define the loop defect

$$H_F(\gamma) = T_\gamma - I.$$

Definition 9.1 (Operational holonomy candidate). A reproducible path-dependent family $\gamma \mapsto T_\gamma$ on closed intervention loops is an operational holonomy candidate. This name indicates the algebraic role, not the prior existence of an affine or principal connection.

Proposition 9.2 (Gauge-relative reconstruction target). *Even if a forward model maps an admissible connection ∇ to all loop operators, reconstruction can at most target an equivalence class $[\nabla]$ whenever gauge-related connections have identical loop data. A connection theorem therefore requires a specified bundle or transport object, a path family, a gauge relation, and injectivity of the forward map modulo gauge.*

K7 firewall

A memory kernel acts on histories. A connection acts on sections or fibres. A loop response is an observable operator. These are different types. The arrow $K \rightarrow [\nabla]$ is not meaningful until a forward operator $\mathcal{A} : [\nabla] \mapsto K$ or $\mathcal{A} : [\nabla] \mapsto \{T_\gamma\}$ is declared and shown injective on the admitted class.

10 The final theorem and corollary ladder

Theorem 10.1 (Resolution Geometry closure theorem). *Let $F : X \rightarrow Y$ be an observation map.*

- (i) *F forces the exact equivalence relation R_F , quotient Q_F , and universal factorization of fibre-constant constructions.*
- (ii) *A declared differentiable lift and positive-definite observation covariance force the pullback form g_F and $\text{rad } g_F = \ker dF$.*
- (iii) *Constant rank identifies this radical with the exact fibre tangent. Without that gate, finite-scale separation can falsify exact nullness, but unrestricted smooth numerical or jet data cannot certify it.*
- (iv) *A projectable intervention-response assignment satisfying identity and serial composition defines a functor from the resolved intervention category to observable response operators.*
- (v) *A positive closure defect is a witness that the selected resolved representation does not close for the tested composition.*
- (vi) *No causal order, memory kernel, affine connection, or conformal metric is forced by static relict data or by a closure defect alone.*
- (vii) *An observable causal cone can be reconstructed only after a directed intervention-response relation is declared and measured, descends to Q_F , is stable under the resolution protocol, and passes the required causal, topological, and quadratic-cone gates.*

Corollary 10.2 (Forced and gated chain). *The supported chain is*

$$F \rightarrow R_F \rightarrow Q_F \quad \text{and, after declared structure,} \quad (F, \Sigma) \rightarrow g_F \rightarrow \ker dF,$$

together with the independent operational chain

$$Q_F \rightarrow \mathcal{I} / \sim_F \rightarrow T \rightarrow \Delta_F.$$

The branches

$$\Delta_F \rightsquigarrow \preceq, \quad \preceq \rightsquigarrow [g], \quad \{T_\gamma\} \rightsquigarrow [\nabla], \quad Y \rightsquigarrow K$$

are conditional inverse problems and require their own forward definitions, projectability, completeness, and identifiability theorems.

Corollary 10.3 (Operational transport obstruction). *Exact observable transport closure on the declared quotient requires*

$$\Delta_F(a, b) = 0$$

for every admitted composable pair. Therefore a persistent, statistically resolved positive defect obstructs an exact functorial transport model on the chosen resolved state. The defect localizes a closure failure but does not uniquely identify its physical cause.

Corollary 10.4 (Light-cone computation criterion). *A conformal light cone is computable from experiments only if the directed response protocol yields a projectable and stable local reachability cone, the cone boundary is compatible with a Lorentzian quadratic form, and the hypotheses of the selected causal realization theorem hold. The output is $[g]$ and an orientation supplied by the directed intervention protocol; absolute scale requires additional calibration.*

11 Attack batteries and reproducibility record

The batteries are evidence about implementations and countermodels. They are not substituted for analytic proof.

Battery	Seeds	Scope	Recorded outcome
Pullback and relict attack	1729, 271828, 314159	Six symbolic and numerical attacks	Same kernel pair with different pullbacks; singular fibre mismatch; finite-jet and smooth-flat attacks; covariance nonidentifiability; coordinate covariance; one-sided multiscale witness. All six registered checks passed.
Prediction-lock attack	1729, 271828, 314159	Five proposed extrapolations plus locked controls	Zero unconditional extrapolations survived. Three mathematical results locked and one conditional empirical target survived.
K7 direction attack	1729, 271828, 314159	Nine directions from ρ or response toward dynamics and geometry	No unconditional physical reconstruction survived. Conditional interfaces remained for intervention support to preorder, causal order to conformal class, and gauge-relative holonomy reconstruction. Composable response representation was the minimal positive route.
Final K2–K3 and K7-I battery	20260912	300 pullback coordinate trials; rank-deficient radical check; 200 pairs in each block-linear coupling regime	Maximum covariance error 1.68×10^{-12} ; kernel quadratic value 2.00×10^{-31} ; mean $\Delta_F = 0.0000$ in the decoupled regime; mean $\Delta_F = 0.2394$ at coupling amplitude 0.6.

11.1 Registered implementation details

The final battery used $J \in \mathbb{R}^{4 \times 3}$, positive-definite $\Sigma = BB^\top + I_4$, invertible sampled observation and parameter transforms, and a rank-two Jacobian for the radical check. The K7-I model used $d_r = 3$, $d_m = 2$, independent Gaussian block matrices, a decoupled regime with cross blocks zero, and a coupled regime in which cross blocks were multiplied by 0.6.

Remark 11.1 (Interpretation boundary). The value 0.2394 is not a universal constant and is not a prediction of nature. It is the mean normalized defect of the stated synthetic model under the stated seed and sampling protocol. Its purpose is to demonstrate that the proposed statistic responds to a controlled hidden write-read channel.

11.2 Reproduction checklist

1. Record Python and NumPy versions and deterministic random seeds.
2. Run the symbolic counterexamples independently of floating-point rank tests.
3. Preserve the normalization in Definition 6.1 when comparing defect means.
4. Report all attempted pairs, not only positive defects.
5. For empirical use, add uncertainty intervals, repeated preparations, representative tests, and held-out compositions.
6. For cone reconstruction, retain raw directional scores and rejected nonquadratic fits.

12 Strict non-claims

Non-claim 12.1. The paper does not derive a response, memory kernel, criticality precursor, causal order, affine connection, or spacetime metric from ρ or g_F .

Non-claim 12.2. A finite nonzero ρ witnesses local distinguishability, but a finite zero measurement does not establish exact fibre membership.

Non-claim 12.3. A single zero $\Delta_F(a, b)$ does not prove global dynamic sufficiency. Exact functorial closure requires the identity and composition laws over the admitted intervention category, together with projectability.

Non-claim 12.4. A positive Δ_F does not uniquely identify hidden memory. It proves nonclosure of the selected resolved model. Hidden state, changing dynamics, preparation effects, noise bias, or model misspecification remain alternative causes unless separately controlled.

Non-claim 12.5. An operational reachability cone need not be quadratic. If the quadraticity or Lorentz-signature gate fails, the result remains a directed nonmetric cone or preorder and must not be reported as a Lorentzian light cone.

Non-claim 12.6. A conformal reconstruction yields $[g]$, not an absolute metric scale. A connection reconstruction, if available, is gauge-relative and depends on path completeness and a declared forward map.

Non-claim 12.7. The numerical batteries are reproducibility and counterexample records. They do not prove universal analytic statements by sampling.

13 Discussion: what is genuinely new

The first novelty is the exact location of the type boundaries. The observation quotient, differential null sector, finite-scale profile, response algebra, causal preorder, cone, conformal class, kernel, and connection are not interchangeable descriptions of one object. The ladder prevents information from a higher layer being silently inserted into a lower one.

The second novelty is the operational treatment of dynamic sufficiency. Instead of declaring that unresolved variables carry memory, one compares true serial response with composition of one-step resolved operators. The defect Δ_F is measurable, representative-testable, and independent of any prior metric or connection.

The third novelty is the cone computation firewall. A light cone is not read from Fisher nullness. It is built from directed intervention response, passed through quotient projectability and resolution-stability tests, then fitted to a Lorentzian quadratic cone. Only after these steps does conformal causal reconstruction become applicable.

The fourth novelty is negative but productive. Static relict structure supplies an experimentally useful covariate, not the sign or functional form of a physical response law. This identifies the precise place where new physics must be posted: a falsifiable K6 compatibility law or a forward K7 reconstruction map.

14 Conclusion

The foundation closes at a smaller and stronger object than a derived spacetime. The exact observation map forces its kernel pair and quotient. A declared differentiable statistical lift forces a pullback form whose radical is the differential null sector. Regularity identifies that sector with the exact fibre tangent; singular finite-scale response can falsify exact nullness but cannot certify it numerically in unrestricted smooth models.

The positive bridge beyond local geometry is composability. Projectable response operators form a functor exactly when the declared resolved representation obeys identity and serial composition. The closure defect is the obstruction. In the executed block-linear attack, the defect vanished when the null sector was decoupled and became positive when a hidden write-read path carried two-step memory.

Causal cones and transport geometry remain attainable, but only conditionally. Directed response support can generate an operational preorder. Stable projectable local reachability can generate a cone candidate. A Lorentzian quadratic gate and causal reconstruction hypotheses can then yield a conformal class. Loop responses can become holonomy data only after a path and gauge framework is declared. These are explicit reconstruction problems, not consequences of static Fisher or relict data.

The final Resolution Geometry foundation is therefore a forced chain, a family of gates, and an auditable set of obstructions. It states what observation fixes, what finite resolution can witness, when resolved dynamics close, and exactly what additional evidence is required before the words memory, causality, light cone, connection, or spacetime metric are justified.

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A Minimal executable battery

```
import numpy as np
rng = np.random.default_rng(20260912)

# K2-K3 coordinate covariance
maxerr = 0.0
for _ in range(300):
    J = rng.normal(size=(4,3))
    B = rng.normal(size=(4,4))
    S = B @ B.T + np.eye(4)
    A = rng.normal(size=(4,4))
    C = rng.normal(size=(3,3))
    if abs(np.linalg.det(A)) < .1 or abs(np.linalg.det(C)) < .1:
        continue
    g = J.T @ np.linalg.inv(S) @ J
    gp = (A@J@C).T @ np.linalg.inv(A@S@A.T) @ (A@J@C)
    target = C.T @ g @ C
    maxerr = max(maxerr, np.linalg.norm(gp-target)/(1+np.linalg.norm(target)))

# K7-I block-linear closure defect
DR, DM = 3, 2
def intervention(coupling):
    R = rng.normal(size=(DR,DR))
    M = rng.normal(size=(DM,DM))
    U = coupling*rng.normal(size=(DR,DM))
    V = coupling*rng.normal(size=(DM,DR))
    return np.block([[R,U],[V,M]])

def defect(B,A):
    true = (B@A)[:DR,:DR]
    naive = B[:DR,:DR] @ A[:DR,:DR]
    return np.linalg.norm(true-naive)/(1+np.linalg.norm(true))
```

B Cone-fit design matrix

For n dimensions, write a symmetric matrix G using $n(n+1)/2$ independent coefficients. For each boundary ray $u = (u_1, \dots, u_n)$, the equation $u^\top G u = 0$ is linear in those coefficients. In three dimensions one row may be written

$$(u_1^2, u_2^2, u_3^2, 2u_1u_2, 2u_1u_3, 2u_2u_3).$$

The smallest right singular vector gives a candidate coefficient ray. This linear fit does not itself establish Lorentzian geometry. Signature, held-out error, interior/exterior sign consistency, projectability, causal orientation, and stability remain mandatory gates.